BACSE203: COMPUTER NETWORKS — CASE STUDY 5

URGENT 5-PERSON WORK SPLIT (SUBMISSION TOMORROW)

Topic: POP3 Mailbox Access in Low-Bandwidth Branch • **Action Plan:** Copy-paste ready task allocation.

PERSON 1 (LEAD) — PROTOCOL & QUESTION (a)

Owns: Section 1 (Overview) & Section 6.1 (Question a: Explain POP3 and Its Limitations).

What to write:

1. Explain POP3 3-state lifecycle: AUTHORIZATION (USER, PASS), TRANSACTION (STAT, LIST, RETR, DELE, TOP, UIDL), UPDATE (QUIT).
2. Explain why it fails on low bandwidth: Monolithic download (downloads whole email + heavy attachment together).
3. Explain lack of resume: If link drops at 90%, it restarts from byte 0 (no byte-range offset in RETR).
4. Explain non-atomic delete: Deletions only happen on clean QUIT. Link drop rolls back deletes, causing duplicate emails.

PERSON 2 — WORKFLOW DESIGN & QUESTION (b)

Owns: Section 1.1–1.4 (WAN Setup) & Section 6.2 (Question b: Design Email Workflow).

What to write:

1. Setup parameters: 256 kbps bandwidth, 450 ms satellite RTT, 5% packet loss.
2. 5-Tier Resilient Workflow:

• Tier 1: Local branch cache/proxy (Dovecot/Fetchmail) on LAN so users get instant local access.
• Tier 2: Two-Phase Header Pre-fetch: Use TOP msg 0 to fetch sender, subject, size (<15 KB for 50 emails).
• Tier 3: Attachment Throttling: Auto-download <100 KB; hold >100 KB at HQ until user clicks or overnight.
• Tier 4: Compression: Gzip/zlib compression on the WAN link (cuts text payload by 65%).
• Tier 5: Outbound Spooling: Local Postfix relay queuing outbound mail with exponential retry backoff.

PERSON 3 — SIMULATION & PACKET CAPTURE (SECTION 5)

Owns: Section 5 (Implementation, Simulation & Demonstration) & Table 4.

What to write:

1. Fill Table 4: Tools used (Wireshark 4.2.x, Python 3.12 socket, Ubuntu Linux tc-netem for 256 kbps / 450 ms latency).
2. Wireshark Packet Trace: Show frames for TCP 3-way handshake, plaintext USER/PASS on port 110, RETR streaming.
3. Proof of Lockout: Show Frame 86 disconnect (TCP RTO timeout) and Frame 94 reconnection error: -ERR [IN-USE] Mailbox is locked.
4. Python Socket Test: Show code snippet extracting headers via TOP 4 0 in 820 bytes without downloading 1.88 MB.

PERSON 4 — SECURITY & QUESTION (c)

Owns: Section 6.3 (Question c: Security and Reliability Concerns).

What to write:

1. Security Vulnerabilities: Cleartext sniffing on port 110, MITM payload tampering, lack of MFA/OAuth.
2. Cryptographic flaw of APOP: RFC 1939 uses MD5; explain why MD5 collision & length-extension attacks break it.
3. Reliability failures: Orphaned TCP sessions keeping dotlock/flock locked for 15-30 min; corrupted PST/mbox files.
4. Hardening Blueprint: Enforce POP3S on port 995 with TLS 1.3 (TLS_AES_256_GCM_SHA384), SASL XOAUTH2, DMZ firewall.

PERSON 5 — TABLES, BENCHMARKS, REFERENCES & POLISH

Owns: Table 2, Table 3, Table 5, Section 7 (Conclusion), Section 8 (References).

What to write:

1. Fill Table 2 (Protocols): POP3, IMAP4, SMTP, POP3S, MIME (purpose, features, advantages, limits).
2. Fill Table 3 (RFCs): RFC 1939 (POP3), RFC 2595 (TLS), RFC 3501 (IMAP), RFC 5321 (SMTP) with URLs & access dates.
3. Calculate Base64 overhead: 3 binary bytes = 4 ASCII chars (33.33% + CRLF wrap = 37% inflation). 3 MB file becomes 4.11 MB.
4. Performance Table 5: Compare Legacy POP3 vs IMAP4 vs Proposed Edge Proxy (118s vs 42s vs 1.9s).
5. Format references & final document check in Word.